

Participant-Facing Materials

Recruitment, screening, consent-information, session script, and debrief drafts

FIRST-PASS DRAFT

Not IRB-approved. Replace all bracketed open decisions and transfer final content into the current official UCF forms before submission.

A. Recruitment Email

Subject: Mixed-Reality Visual Search Study Recruitment

Researchers at the University of Central Florida are inviting adults to participate in a study about mixed-reality object search and computer guidance.

Participation involves one in-person session lasting approximately [OPEN DECISION: duration]. You will wear an HTC Vive Focus Vision headset with eye tracking, search for virtual objects on a shelf-like display, hear spoken guidance, and answer questionnaires. You may experience temporary eye strain, headset discomfort, fatigue, dizziness, nausea, or frustration.

- You must be at least 18 years old.
- You must be able to understand the approved English instructions and consent information.
- You must be able to safely wear a mixed-reality headset, see the task colors and objects with normal or corrected vision, and hear spoken guidance.
- You should not participate if you currently feel dizzy, nauseous, unwell, or unusually fatigued, or if you have a condition that makes headset use unsafe.

Compensation: [OPEN DECISION]. Participation is voluntary, and choosing not to participate will not affect your grades, employment, services, or relationship with UCF.

Study contact: [OPEN DECISION: approved PI name and contact]

Scheduling link or QR code: [OPEN DECISION]

B. Minimal Eligibility Prescreen

ADMINISTRATION NOTE

Collect only what is needed to determine eligibility before consent. Do not retain medical detail or research questionnaire responses from people who do not consent unless the approved protocol explicitly permits it.

1. Are you 18 years of age or older?

Yes / No

2. Can you read and understand the approved English consent and task instructions?

Yes / No

3. Can you safely wear a mixed-reality headset for approximately [duration] with breaks?

Yes / No / Unsure

4. Do you currently feel dizzy, nauseous, unwell, have a severe headache, or feel unusually fatigued?

Yes / No / Prefer not to say

5. Do you have a history of severe simulator sickness, a relevant vestibular disorder, or visually triggered seizures that would make headset use unsafe?

Yes / No / Prefer not to say

6. With your usual correction, can you see and distinguish the task objects/colors and hear spoken guidance?

Yes / No / Unsure

7. Are you willing to allow coded gaze, interaction, task-performance, and questionnaire data collection?

Yes / No

Eligibility outcome: Eligible / Not eligible / PI review required. Record only the outcome and approved minimal screening fields.

C. Participant Information and Consent Draft

TEMPLATE NOTE

This is content for review, not a substitute for the current official UCF consent template. The PI and IRB must determine written versus verbal consent.

Study title: Gaze-Contingent Guidance in Mixed-Reality Visual Search

Research team: [OPEN DECISION: PI, faculty advisor, department, contacts]

Why am I being invited?

You are being invited because you are an adult who may be able to safely use a mixed-reality headset and complete a visual-search task.

Why is this research being done?

The purpose is to understand whether spoken computer guidance that uses information about where a person is looking changes object-search performance, workload, and experience compared with guidance that does not use gaze information.

What will happen?

If you agree, you will complete brief background and safety questions, wear an HTC Vive Focus Vision mixed-reality headset, complete eye-tracking calibration and practice, and search for announced target objects among virtual distractors. You select an object by looking at it continuously until the dwell indicator completes. The system may provide spoken guidance. Some guidance may use current or recent gaze behavior, while other guidance does not. You may complete workload, sickness, usability, and experience questions and a short interview. The visit will last approximately [OPEN DECISION].

What information will be collected?

The study may collect a coded participant ID, task condition and configuration, gaze origin and direction, available eye openness and device fixation-point signals, object hover and dwell behavior, target and selected object metadata, timing, wrong and correct selections, estimated blinks when available, spoken-guidance events, calibration/quality information, questionnaire responses, and researcher-recorded technical deviations. The current study does not intend to record your face, room video, or ordinary conversation.

What are the risks?

Possible temporary risks include eye strain, headache, fatigue, warmth or pressure from the headset, dizziness, nausea, disorientation, frustration, and discomfort about eye-tracking privacy. You may pause, request a break, skip optional questions, or stop at any time. Tell the researcher immediately if you feel uncomfortable.

Are there benefits?

You may not receive a direct personal benefit. The research may contribute to the design of mixed-reality systems that provide better-timed or more relevant assistance.

How will privacy be protected?

Research files will use a coded participant ID. Identifying scheduling, consent, and compensation information will be stored separately from research data. Access will be limited to approved study personnel. Data will be stored on [OPEN DECISION: approved UCF-managed storage] for [OPEN DECISION: retention period] and then handled according to the approved deletion or archiving plan. Complete secrecy cannot be guaranteed; authorized UCF or IRB representatives may inspect study records.

Do I have to participate?

No. Participation is voluntary. You may decline, stop, or withdraw without penalty or loss of benefits to which you are otherwise entitled. The approved form will explain whether and until when already-collected data can be removed.

Compensation and contacts

Compensation: [OPEN DECISION: amount, form, completion and partial-payment policy]. Study questions: [PI/faculty advisor contacts]. Questions about your rights as a research participant: [current UCF IRB contact language from the official template].

D. Standardized Session Script

1. Welcome the participant, confirm eligibility, review consent, answer questions, and obtain approved consent.
2. Assign the coded ID and begin the session manifest.
3. Administer baseline questionnaires and baseline SSQ.
4. Explain that breaks and stopping are always available and that wrong selections are expected.
5. Fit the headset, adjust for comfort, and complete vendor eye calibration.
6. Validate gaze across the search area. Refit and recalibrate if the approved criterion is not met.
7. Explain: locate the announced color-shape target as quickly and accurately as possible; select by continuous gaze dwell; continue after a wrong selection; spoken hints may be used or ignored.
8. Run practice until the approved criterion is met without experimenter intervention.
9. Run the assigned condition schedule. Do not reveal condition labels or hypotheses.
10. At approved breaks, check comfort using neutral language and administer scheduled post-block measures.
11. After the task, administer post-SSQ, SUS, comparative questions, and interview before revealing the manipulation.
12. Remove the headset, check comfort, debrief, confirm compensation, and record deviations or withdrawal requests.

E. Stopping and Adverse-Event Script

Stop the task immediately if the participant asks to stop or reports significant nausea, dizziness, disorientation, headache, eye strain, distress, or other concerning symptoms. Pause the application, assist with safe headset removal, offer a seated rest, and follow the approved escalation/reporting procedure. Do not encourage the participant to continue. Record the coded event and only the minimum approved detail.

F. Debriefing Script

Thank you for participating. This study compares spoken guidance that can use gaze information with guidance that does not use gaze information. In gaze-aware conditions, the assistant could use where and how you were looking to choose feedback. In gaze-unaware conditions, it used general guidance without gaze evidence. We are interested in whether this changes search performance, workload, trust, helpfulness, or privacy perceptions. Your coded data will be handled as described in the consent information. Please contact the research team if you have questions or wish to request withdrawal within the approved timeframe.